Sahaj Bhadja

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Summary

Software engineer completing an MSCS at USC (available May 2026) with internship experience building impact-driven, AI-powered web products in **TypeScript**, **React**, **Node.js**, and **Python**. Passionate about full-stack features—from polished UI to **PostgreSQL** data models—that improve performance, reliability, and real user workflows.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4.0*
Coursework: Web Technologies, Advanced NLP, Database Systems, Software Engineering

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4.0*
Coursework: Software Engineering, OOP, Data Structures, Operating Systems

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Planned and built AI workspace features end-to-end, architecting async **LLM** orchestration and multi-tenant **PostgreSQL** schemas that support **500+ users**, **1K+ AI jobs**, and **99%+ production reliability** across collaborative education workflows.
- Improved performance and reliability by optimizing indexed queries and dashboard data-fetch patterns, reducing latency from **1.2s** to **350ms** while owning releases, blockers, and cross-functional implementation from conception through maintenance.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Shipped full-stack product features from research to maintenance using **React**, **Python**/Flask, and **MySQL**, integrating multimodal **LLMs** and OCR into onboarding-style upload, rubric, and grading flows serving **10K+ daily requests**.
- Owned outcomes independently with clear progress communication, building internal automation and **GitHub Actions CI/CD** pipelines that reduced instructor feedback turnaround by **60%** while collaborating with design-minded product workflows.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built AI-assisted pipelines processing **100K+ trajectory logs**, applying transformer-based modeling and thoughtful problem decomposition to improve long-horizon planning stability by **15%** through iterative research and clean implementation.
- Partnered cross-functionally with researchers on data tooling and evaluation workflows, demonstrating initiative in identifying bottlenecks, documenting decisions, and delivering maintainable systems others could build on.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Improved reliability of distributed platform services through **20+ automated test suites**, nightly integration workflows, and performance tuning that increased pipeline efficiency by **68%** and API response times by **30%**.
- Participated in code reviews and architectural discussions with Microsoft/GitHub partners, troubleshooting production issues across **Linux**, Windows, and macOS while writing maintainable, impact-focused backend modules.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Built and maintained quality **React.js** and **Node.js** interfaces integrated with **REST APIs** and **SQL** backends across a microservices CRM platform, syncing **2M+ records** with real-time updates for operations teams.
- Contributed to internal tooling and deployment automation with **Docker** and Jenkins, cutting release cycles by **50%** while polishing existing features and supporting cross-functional delivery in a fast-paced environment.

Skills

Web & Product: TypeScript, JavaScript, React.js, Node.js, Next.js, HTML, CSS, Python, FastAPI, Flask, full stack development, web applications, product features, user impact, ownership, cross-functional collaboration, problem solving
Data & AI: PostgreSQL, MySQL, MongoDB, SQL, Elasticsearch (familiar), REST APIs, LLMs, AI-assisted development, performance optimization, reliability, Git, GitHub Actions, CI/CD, Docker, Agile, internal tools

Projects

FinNexus Live | *TypeScript, Next.js, MongoDB, Real-Time Web*

- Built a scalable full-stack fintech product with **TypeScript**, secure APIs, and real-time transaction updates across **1M+ records**, demonstrating end-to-end ownership from UI polish to backend data models and deployment automation.

HMIS Live | *TypeScript, React, Workflow UI, Product Features*

- Designed user-facing workflow features with **TypeScript** and **React**, integrating appointment, notification, and record-access flows that reduced administrative workload by **30%** for **100+ daily** operational users.

Linguistix | *Python, Node.js-Ready APIs, Docker, CI/CD*

- Developed an AI-ready document ingestion API with containerized deployment and **CI/CD** releases, building internal-tooling patterns that improved retrieval workflows and made backend services easier for frontend teams to integrate.